

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 2: Communication

Subtopic 2.1: Networks including the Internet

Concept 2.1.3: Thin-Client vs Thick-Client

Total Questions: 3

Mark Schemes begin on Page 4

Question 1 | Source: 9618_s24_qp_12 (Q3.b)

3 An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(b) The exam markers use software that operates as a thin-client to mark the exam papers.

Complete the table by identifying **two** characteristics of a thin-client.

Describe how each characteristic will be used in this software.

	Thin-client characteristic	Description of use in this software
1		
2		

[4]

Question 2 | Source: 9618_s23_qp_12 (Q1.e)

1 A company has a LAN (local area network).

(e) The network runs as a thick-client model.

Describe what is meant by a **thick-client** model.

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..... [2]

Question 3 | Source: 9618_s23_qp_13 (Q2.c)

2 A university has two sites. Each site has several computer rooms. The computers are all connected as a WAN (wide area network).

(c) The computers in one room are set up as thin-clients in a client-server model.

Describe the role of the different computers in this model.

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..... [2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_s24_ms_12 (Q3.b)

3(b)	1 mark for characteristic 1 mark for description of application to examination software:	4												
<table><tr><th>Thin-client characteristic</th><th>Description of use in this software</th></tr><tr><td>Data is not stored on the client computer</td><td>Exam papers are stored on the server and not on the examiner's computer // exam papers are not permanently stored on the examiners' computers</td></tr><tr><td>Client computer is reliant on access to server</td><td>Examiners cannot mark if their device cannot access the server / the server 'goes down'</td></tr><tr><td>Client computer heavily reliant on network/internet connection</td><td>The marking software will not operate without network/internet access</td></tr><tr><td>Client computer requires few local resources/memory</td><td>Examiners can use devices with low resources and the marking software will still function</td></tr><tr><td>Client computer performs minimal functions/processes</td><td>The marking software transmits requests, the server responds and sends the response to the user</td></tr></table>			Thin-client characteristic	Description of use in this software	Data is not stored on the client computer	Exam papers are stored on the server and not on the examiner's computer // exam papers are not permanently stored on the examiners' computers	Client computer is reliant on access to server	Examiners cannot mark if their device cannot access the server / the server 'goes down'	Client computer heavily reliant on network/internet connection	The marking software will not operate without network/internet access	Client computer requires few local resources/memory	Examiners can use devices with low resources and the marking software will still function	Client computer performs minimal functions/processes	The marking software transmits requests, the server responds and sends the response to the user
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Mark Scheme for Question 2 | Source: 9618_s23_ms_12 (Q1.e)

1(e)	1 mark each - The server performs minimal / some processing for the client - The clients also do most of their own processing/work independently // most of the resources are installed locally	2
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Mark Scheme for Question 3 | Source: 9618_s23_ms_13 (Q2.c)

2(c)	1 mark each to max 2 - Server performs all processes required by the task and/or data storage - Clients only sends requests to the server and displays the returned results	2
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